

Assignment Answers

1. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every task. By analyzing patterns in data, machine learning models can improve their performance over time and provide accurate results for various real-world applications such as recommendation systems, spam detection, and weather forecasting.

2. What is a Feature?

A feature is an input variable or attribute used by a machine learning model to make predictions. Features provide information about the data and help the model identify patterns. For example, in a student performance dataset, the number of hours studied can be considered a feature because it influences the student's exam score.

3. What is a Label?

A label is the output variable or target value that the machine learning model is trying to predict. It represents the correct answer in the dataset. For example, if the goal is to predict a student's exam score based on study hours, then the exam score is the label.

4. Why do we split data into Training and Testing Sets?

Data is split into training and testing sets to evaluate the performance of a machine learning model. The training set is used to teach the model by identifying patterns in the data, while the testing set is used to check how well the model performs on unseen data. This helps ensure that the model can generalize well and is not simply memorizing the training data.

5. What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values. It establishes a relationship between input features and the target variable by fitting a straight line through the data points. The algorithm is widely used because it is simple, easy to understand, and effective for predicting values such as prices, scores, and sales based on historical data.